

LINE 2 HOT FILLER

Printable System Review Packet - HMI Inventory, Process Relationships, Controls, Field Verification and Reference Literature

Review status: WORKING DOCUMENT - HMI-derived information and field observations. It is not an as-built P&ID; or validated PLC program. Anything inferred is confidence-marked and should be field-verified before being treated as authoritative.

SYSTEM	Line 2 HOT FILLER
FACILITY	J. Lieb Foods / facility context from supplied HMI material
PRIMARY REVIEW AREAS	Raw product, product heating, hold, cooling, return, filler, CIP/water, steam, condensate, electrical/pneumatic control
CURRENT PRIORITY	TV101 steam/heating behavior and physical heat-delivery chain
FIELD DEVICE IDENTIFIED	Fairchild TT6000-401 electro-pneumatic transducer
CONFIDENCE KEY	CONFIRMED = directly observed; LIKELY = strong context/standard usage; BEST GUESS = useful working interpretation; UNKNOWN = unresolved
REVIEWER	_____
DATE	_____

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1. REVIEW METHOD

Use the HMI snapshot as a point-in-time operating record. Keep the tag identity separate from its temporary reading. Example: **PMP104** is the device; **49 Hz** is the snapshot value. A relationship such as PMP104 CIRCULATES heating water is separately confidence-marked.

MARK	MEANING	HOW TO USE IT
CONFIRMED	Directly visible on HMI/nameplate or otherwise established	May be used as a working fact; still verify before hazardous work.
LIKELY	Strongly supported by HMI placement, standard instrumentation usage, or physical context	Use for mapping; field-check piping/tubing/wiring.
BEST GUESS	Plausible expansion/function but not established	Keep in graph so nothing is lost; actively verify.
UNKNOWN	Not enough evidence	Do not expand or invent. Record exact tag and investigate.

Important correction rule: Never invent PLC I/O addresses, ladder rungs, interlocks, valve fail action, or exact pipe routing from appearance alone. Any earlier conceptual ladder drawing containing addresses should be treated as illustrative only, not machine logic.

2. HMI SNAPSHOT - DEVICES, SENSORS AND READINGS

These values are from the working HMI snapshot already extracted for this review packet. They are not design setpoints unless specifically identified as such.

TAG	WORKING TYPE / MEANING	SNAPSHOT	ASSOCIATION	CONF.
DP1	Differential pressure indication	2 PSI	Process differential pressure	LIKELY
DP2	Differential pressure indication	9 PSI	Process differential pressure	LIKELY
DP3	Differential pressure indication	1 PSI	Process differential pressure	LIKELY
PT101	Pressure transmitter	0 PSI	Product line	LIKELY
PT102	Pressure transmitter	2 PSI	Hold/product section	LIKELY
PT103	Pressure transmitter	34 PSI	Heating loop area	LIKELY
PT103A	Pressure transmitter / indication	0 PSI	Upper heating area	LIKELY
PT104	Pressure transmitter	9 PSI	Cooling/process section	LIKELY
PT105	Pressure transmitter	1 PSI	Upper process section	LIKELY
PT106	Pressure transmitter	1 PSI	Process/water section	LIKELY
PT107	Pressure transmitter	0 PSI	Process/water section	LIKELY
TT101	Temperature transmitter	185.2 F	Hold/product temperature	LIKELY
TT102	Temperature transmitter	184.5 F	Heating/cooling process	LIKELY
TT103	Temperature transmitter	188.8 F	Upper heating/process	LIKELY
RTD103	Resistance temperature detector	182.2 F	Process/return area	LIKELY
RTD105	Resistance temperature detector	140.9 F	Water/process area	LIKELY
RTD310	Resistance temperature detector	107.3 F	Balance tank association	LIKELY
FM101	Flow meter	19.5 G/M	Process/product flow	LIKELY
LT301	Level transmitter	109 GAL	150 GAL balance tank	LIKELY
LT302	Level transmitter	21 GAL	Collection tank	LIKELY
PMP105	Raw product supply pump	0 Hz	RAW PROD SUPPLY	CONFIRMED FUNCTION
PMP101	Pump	0 Hz	Product/process line	CONFIRMED TYPE
PMP102	Pump	29 Hz; PID nearby	Process/product	CONFIRMED TYPE
PMP103	Pump	16 Hz; PID nearby	Return/collection side	CONFIRMED TYPE
PMP104	Pump	49 Hz	Heating circulation association	CONFIRMED TYPE
TV101	Modulating control valve	PID; ~5%	Steam/heating control	LIKELY FUNCTION
TV102	Modulating control valve	PID; 0%	Cooling water	LIKELY FUNCTION
TV103	Modulating control valve	PID; 100%	Process/water	LIKELY FUNCTION
TV104	Modulating control valve	PID; 10%	Process/water	LIKELY FUNCTION
TV108	Modulating control valve	PID; 100%	Heating/cooling section	LIKELY FUNCTION
AG301	Agitator	-	Balance tank	LIKELY
AV102/103/104/105/106	Actuated/automatic valves	-	Product/process/collection areas	LIKELY
AV107/108/109/110/111/113	Actuated/automatic valves	-	Water/cooling areas	LIKELY
AV112	Actuated/automatic valve	-	Steam area	LIKELY
FP302	Facility process designation	-	Process path	UNKNOWN
SRV101	Valve designation	-	Product/process path	BEST GUESS
MVP101	Modulating/process valve designation	-	H2O/process supply	BEST GUESS
CP301	Facility control/process designation	0.0%	Process section	UNKNOWN

TAG	WORKING TYPE / MEANING	SNAPSHOT	ASSOCIATION	CONF.
BT1	Tank/batch indication	317 GAL	HMI tank indication	BEST GUESS EXPANSION
BT2	Tank/batch indication	3034 GAL	HMI tank indication	BEST GUESS EXPANSION

3. ACRONYM / TAG DICTIONARY

TAG	WORKING EXPANSION	CONFIDENCE
PMP	Pump	CONFIRMED
PT	Pressure Transmitter	LIKELY
TT	Temperature Transmitter	LIKELY
RTD	Resistance Temperature Detector	CONFIRMED standard term
LT	Level Transmitter	LIKELY
FM	Flow Meter	LIKELY
DP	Differential Pressure	LIKELY
AV	Automatic / Actuated Valve	LIKELY
TV	Control valve; Temperature Valve is a possible facility expansion	BEST GUESS expansion / LIKELY function
AG	Agitator	LIKELY
BT	Batch/Balance Tank or related tank designation	BEST GUESS
MVP	Modulating Valve / process valve designation	BEST GUESS
CP	Control / Process designation	UNKNOWN
FP	Flow / Process designation	UNKNOWN
SRV	Sanitary / Return / Relief Valve possible	BEST GUESS
PID	Proportional-Integral-Derivative	CONFIRMED
CIP	Clean-In-Place	CONFIRMED
RTN	Return	CONFIRMED
COND.	Condensate	CONFIRMED
PROD	Product	CONFIRMED
H2O	Water	CONFIRMED
Hz	Hertz	CONFIRMED
PSI / PSIG	Pounds per square inch / gauge	CONFIRMED
GAL	Gallons	CONFIRMED
G/M	Displayed flow unit; likely gallons/minute	LIKELY

4. CIRCUIT-BY-CIRCUIT REVIEW

CIRCUIT	KNOWN / VISIBLE ELEMENTS	MANUAL REVIEW QUESTIONS
RAW PRODUCT / PRODUCT	RAW PROD SUPPLY, PMP105, balance tank, AG301, LT301, RTD310, AV102, PMP101, PT101, inline devices, FP302, SRV101, AV103, hold/product path, filler	Where does raw product enter? Which valve establishes source? Which pump hands off to pasteurizer? Mark every tee, check valve, sanitary valve and bypass.
STEAM / HEATING	Steam supply, TV101, AV112, steam-side heat exchanger, COND., PT103/PT103A area, TT103/TT102/TT101 context, PMP104 circulation association	At 100% TV101 command, is 20 mA present? Is I/P output near its commanded end? Does actuator fully stroke? Is steam pressure available? Is condensate leaving exchanger?
COOLING WATER	COOL H2O, TV102, AV111, AV113, AV110, AV109, cooling exchanger connections	Identify supply vs return and normal valve state. Mark utility pressure/temperature and which product section it cools.
H2O / CIP / RETURN	H2O SUPPLY, RAW CIP RTN, MVP101, AV107, AV108, RTD105, CP301, TV103, TV104, PT106, PT107	Separate product piping from CIP/water piping. Mark cross-connections and route-dependent valves.
RETURN / COLLECTION / FILLER	Return cooler, PMP103, AV105, LT302 collection tank, AV106, END PROCESS, FILLER	Identify every condition that routes product to return, collection, drain, or filler. Verify physically.

FIELD MARKUP: Circle anything that does not match the machine. Cross out wrong relationships. Write the actual upstream/downstream device directly beside the row.

5. RELATIONSHIP GRAPH - WORKING EDGE LIST

FROM	RELATIONSHIP	TO	EVIDENCE	CONF.
LT301	MEASURES	Balance tank level	109 GAL	CONFIRMED DISPLAY
RTD310	MEASURES	Balance tank/process temperature	107.3 F	LIKELY
AG301	AGITATES	Balance tank	HMI placement	LIKELY
AV102	CONNECTED_TO	Balance tank/product outlet path	HMI line	LIKELY
PMP105	SUPPLIES	Raw product circuit	RAW PROD SUPPLY; 0 Hz	CONFIRMED FUNCTION
PMP101	PUMPS / DISCHARGES_TO	Product process path	0 Hz	LIKELY EDGE
PT101	MEASURES	Product-line pressure	0 PSI	LIKELY
FM101	MEASURES	Process/product flow	19.5 G/M	LIKELY
TV101	MODULATES	Steam/heating energy	PID; ~5% snapshot	LIKELY
Steam supply	SUPPLIES	TV101 / steam heat exchanger path	HMI circuit	LIKELY
Steam exchanger	TRANSFERS_HEAT_TO	Heating circulation/process	HMI arrangement	LIKELY
COND.	RETURNS_FROM	Steam heat exchanger	Visible condensate path	LIKELY
PMP104	CIRCULATES	Heating loop	49 Hz	LIKELY
TT101	MEASURES	Hold/product temperature	185.2 F	LIKELY
TT102	MEASURES	Process temperature	184.5 F	LIKELY
TT103	MEASURES	Upper heating/process temperature	188.8 F	LIKELY
Fairchild TT6000-401	CONVERTS	4-20 mA to 3-15 PSIG	Nameplate	CONFIRMED
Fairchild pneumatic output	ACTUATES	Associated pneumatic valve actuator	Exact valve not traced	UNVERIFIED

Graph vocabulary: FEEDS, CONNECTED_TO, DISCHARGES_TO, RETURNS_TO, MEASURES, CONTROLS, COMMANDS, ACTUATES, SUPPLIES, DRAINS_TO, CIRCULATES, TRANSFERS_HEAT_TO, ASSOCIATED_WITH, INTERLOCKS_WITH.

6. TV101 STEAM CONTROL - ELECTRICAL / PNEUMATIC / PROCESS CHAIN

STAGE	DEVICE / SIGNAL	KNOWN	MANUAL CHECK
1	HMI / PID - TV101	Normally observed around 5-8%; reported excursions to 100% while process remains too cold	Trend command with TT101/TT102/TT103 and steam-side pressure
2	PLC analog output	Expected control architecture uses 4-20 mA to the I/P; exact PLC channel is not established	Identify AO channel and measure command without breaking loop unless procedure allows
3	Fairchild TT6000-401	Electro-pneumatic I/P; 4-20 mA input; 3-15 PSIG output; terminal-block model	Identify the exact transducer serving TV101 by tracing tubing/wiring
4	Instrument air supply	TT6000-401 nameplate observed: supply 20-120 PSIG	Measure supply under demand; inspect filters/regulators
5	Pneumatic output	Nominal 3-15 PSIG for -401 configuration	Compare output pressure with electrical command
6	Valve actuator / positioner	Pneumatic pressure produces mechanical valve movement	Verify full stroke, linkage, diaphragm, leaks, sticking and fail action
7	TV101 steam valve	Modulates heating steam function in HMI context	Compare command with actual stem/position
8	Steam heat exchanger	Provides heat to process/heating circulation	Check steam inlet, heat transfer, fouling and condensate removal
9	COND. path / steam trap	Removes condensate	Check flooding, restriction, failed trap, blocked return
10	PMP104 / circulation	49 Hz snapshot	Verify actual circulation/flow rather than frequency command only

Working chain: HMI/PID -> PLC analog output -> 4-20 mA -> Fairchild TT6000-401 -> 3-15 PSIG -> pneumatic actuator/positioner -> TV101 -> steam flow -> heat exchanger -> circulation -> product heat transfer.

7. ASSOCIATED LITERATURE - FAIRCHILD TT6000-401 REVIEW NOTES

This section condenses the manufacturer-family information most useful at the cabinet. It is a review aid, not a substitute for the manufacturer's complete IOM.

ITEM	REFERENCE NOTE FOR MANUAL REVIEW
Manufacturer / family	Fairchild Industrial Products - Model T6000 family; observed unit marked TT6000-401.
Device function	Electro-pneumatic transducer: converts a DC electrical input signal into a proportional pneumatic output.
Observed -401 configuration	4-20 mA input; 3-15 PSIG output; observed nameplate supply range 20-120 PSIG.
TT prefix	TT6000 is the terminal-block version in the T6000 family.
Mounting	T6000 literature describes DIN-rail mounting using the 16893 mounting kit; this matches the cabinet installation style being reviewed.
Pneumatic ports	T6000 documentation shows IN and OUT ports and a vent. Unused ports are plugged. The vent/exhaust must not be treated as a process-air connection.
Why air can be heard at the body/back	The T6000 is a pneumatic relay/nozzle-flapper device and has a vent/exhaust path. A small operational bleed/exhaust can be normal; a large continuous leak at steady command should be compared against the IOM and checked for contamination, damaged internals, supply issues, or downstream behavior.
Calibration	ZERO and SPAN adjustments exist. Do not adjust until input current, supply pressure, output pressure, direction of action, and downstream actuator behavior are verified.
Direction	T6000 family literature supports forward/direct and reverse-acting adjustment. Therefore 15 PSIG cannot automatically be called 'valve open' until the actuator/valve action is verified.
Safety	Steam, hot product, instrument air and energized control circuits can injure personnel. Use plant LOTO/process procedures and rated test equipment; do not defeat interlocks for troubleshooting.

Manufacturer literature to keep with this packet: **Fairchild T6000 Electro-Pneumatic Transducer Installation, Operation & Maintenance Instructions**. Search/print the current T6000 IOM from Fairchild/Rotork or an authorized distributor and verify that it covers the exact TT6000-401 revision/date code installed.

8. TV101 / HEATING TROUBLESHOOTING WORKSHEET

Reported symptom: TV101 often sits around 5-8%, then may command 100% while the system is still too cold. At 100% controller output, first prove the physical heat-delivery chain before changing PID tuning.

TEST POINT	NORMAL / EXPECTED RELATIONSHIP	ACTUAL	PASS?	NOTES / NEXT ACTION
HMI TV101 command	0-100% PID output	_____	_____	_____
I/P input current	Approx. 4-20 mA across command range	_____	_____	_____
I/P supply air	Within device requirement; stable under demand	_____	_____	_____
I/P output pressure	Approx. 3-15 PSIG across configured range	_____	_____	_____
Valve stem / position	Moves smoothly to commanded endpoint	_____	_____	_____
Steam upstream pressure	Stable and adequate for process	_____	_____	_____
Steam downstream of valve	Responds to valve movement	_____	_____	_____
Exchanger steam-side temperature	Responds strongly at high command	_____	_____	_____
Condensate removal	No evidence of flooded exchanger / blocked return	_____	_____	_____
PMP104 actual circulation	Flow established, not merely Hz indication	_____	_____	_____
TT103	Compare against command and heat input	_____	_____	_____
TT102	Compare against command and heat input	_____	_____	_____
TT101	Compare against hold/product condition	_____	_____	_____

9. FIELD WALKDOWN - CIRCUIT TRACE

Circuit: _____ Date: _____ Reviewer: _____ HMI state/product: _____

SEQ	TAG / DEVICE	TYPE	READING	UPSTREAM	DOWNSTREAM	ASSOCIATED DEVICE / CONTROL	CONF.	CORRECTION / NOTES
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								

9B. FIELD WALKDOWN - ADDITIONAL CIRCUIT TRACE

Circuit: _____ Date: _____ Reviewer: _____ HMI state/product: _____

SEQ	TAG / DEVICE	TYPE	READING	UPSTREAM	DOWNSTREAM	ASSOCIATED DEVICE / CONTROL	CONF.	CORRECTION / NOTES
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								

10. CORRECTION / VERIFICATION REGISTER

ITEM TO VERIFY	CURRENT WORKING ASSUMPTION	FIELD CORRECTION / EVIDENCE
TV acronym	Control valve; exact expansion TBD	
FP302	Exact expansion/function unresolved	
CP301	Exact expansion/function unresolved	
SRV101	Exact valve designation unresolved	
MVP101	Exact expansion unresolved	
BT1 / BT2	Tank/batch designation; exact meaning TBD	
TV101 I/P association	Fairchild unit association not yet physically traced	
TV101 fail action	Unknown	
PMP104 exact loop	Heating circulation association likely	
Steam trap / condensate device	Exact device/tag not yet established	
Raw product full route	Needs device-by-device field trace	

Additional notes / sketch:
